D. Y. PATIL COLLEGE OF ENGINEERING AND TECHNOLOGY, KOLHAPUR

(An Autonomous Institute)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

A MINI-PROJECT REPORT ON

PYTHON PROGRAMMING (APC COURSE)

VOICERA

A REAL-TIME MULTILINGUAL SPEECH-TO-TEXT AND AUDIO TRANSCRIPTION WEB APPLICATION USING PYTHON

Submitted by

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Academic Year: 2026-27

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CERTIFICATE

This is to certify that the mini-project work entitled "VOICERA: A REAL-TIME MULTILINGUAL SPEECH-TO-TEXT AND AUDIO TRANSCRIPTION WEB APPLICATION USING PYTHON" has been successfully carried out by the following student of the Department of Computer Science and Engineering, D. Y. Patil College of Engineering and Technology, Kolhapur, during the academic year 2026 – 2027:

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The mini-project report is submitted in partial fulfilment of the requirements for the Python Programming (APC) course and has been reviewed and approved for internal assessment.

Date: _____

Place: Kolhapur

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1. Introduction

VOICERA: Real-Time Multilingual Speech-to-Text and Audio Transcription Web Application using Python (Flask)

VOICERA is a real-time web-based speech-to-text and audio processing application developed in Python. The user gives a command by speaking into the microphone or by typing. The command is sent to a Python (Flask) server, which understands the request, performs the action, stores data in an SQLite database, and sends back a reply that is shown on screen and also read aloud. It can tell the time and date, solve arithmetic expressions, manage notes and to-do tasks, tell jokes, and show system information.

Table 1.1: Technologies Used in VOICERA

Component	Technology
Programming language	Python 3
Web framework	Flask (REST API with JSON)
Database	SQLite (built-in sqlite3 module)
Language understanding	Regular expressions (re module), Python ast module for safe calculation
Voice input / output	Browser Web Speech API (speech recognition) and Speech Synthesis
Frontend	HTML, CSS and JavaScript
Testing	Python unittest

2. SYSTEM ARCHITECTURE DIAGRAM

The system follows a client-server architecture with four main parts: the browser frontend, the Flask server, the intent engine, and the SQLite database.

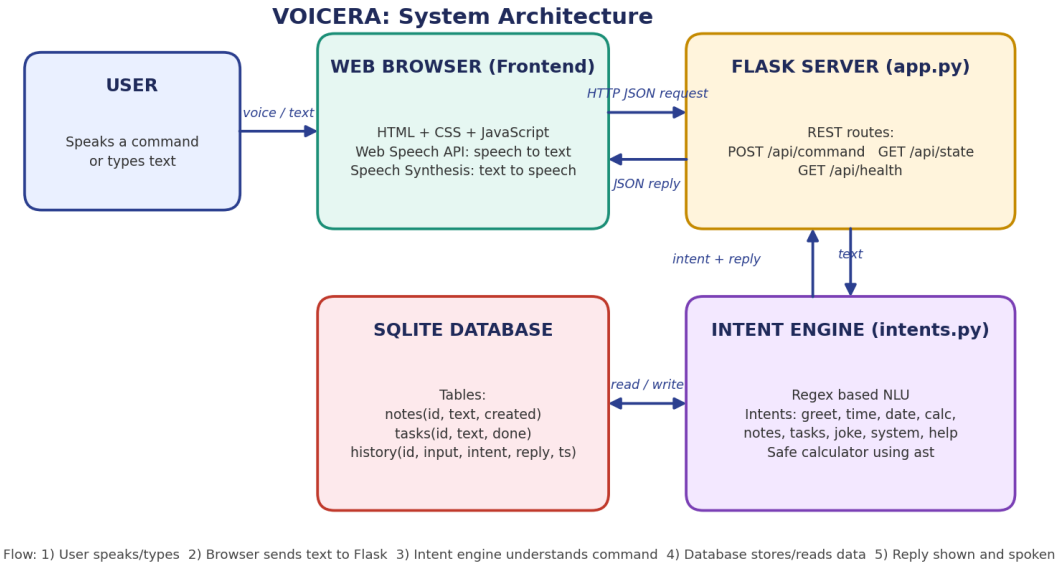


Figure 2.1: System Architecture of VOICERA

2.1 Working of the System

1. User input: The user speaks into the microphone or types a command in the browser.
2. Speech to text: The browser converts speech to text using the Web Speech API and sends it to the server as a JSON request (POST /api/command).
3. Understanding: The Flask route passes the text to the intent engine (intents.py), which matches it against patterns and decides the intent, for example calc, note_add or time.
4. Action and storage: The engine performs the action. Notes, tasks and the command history are saved in or read from the SQLite database.
5. Response: The server returns the intent and reply as JSON. The browser shows the reply in the chat, updates the side panels, and speaks the reply aloud.

3. OBJECTIVES

- To develop a voice-enabled assistant application using Python and the Flask framework.
- To convert spoken or typed commands into actions using a rule-based intent engine written in Python.
- To perform useful tasks such as telling time and date, solving arithmetic expressions, and giving system information.
- To store and manage notes, tasks and command history persistently using an SQLite database.
- To evaluate arithmetic expressions safely using the ast module instead of eval(), preventing code injection.
- To build a simple, responsive web interface with microphone input and spoken replies.
- To test the logic of the application with unit tests written using the unittest module.

4. IMPLEMENTATION: OUTPUT SCREENSHOTS

The application was run locally with the command python app.py. The screenshots below show its output. In the screenshots, commands were entered through the text box; the microphone button gives the same commands by speech in browsers that support the Web Speech API.

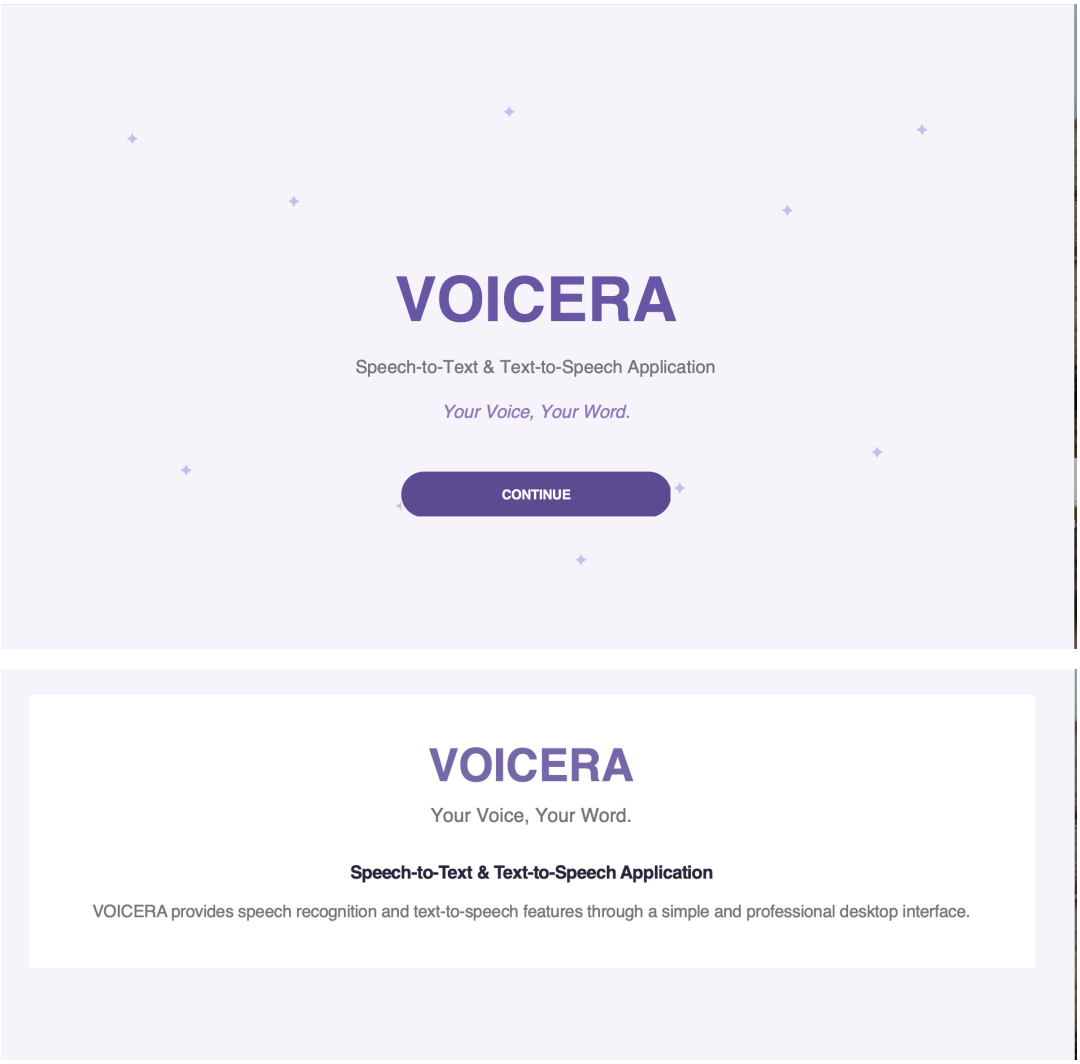


Figure 4.1: Home screen with welcome message

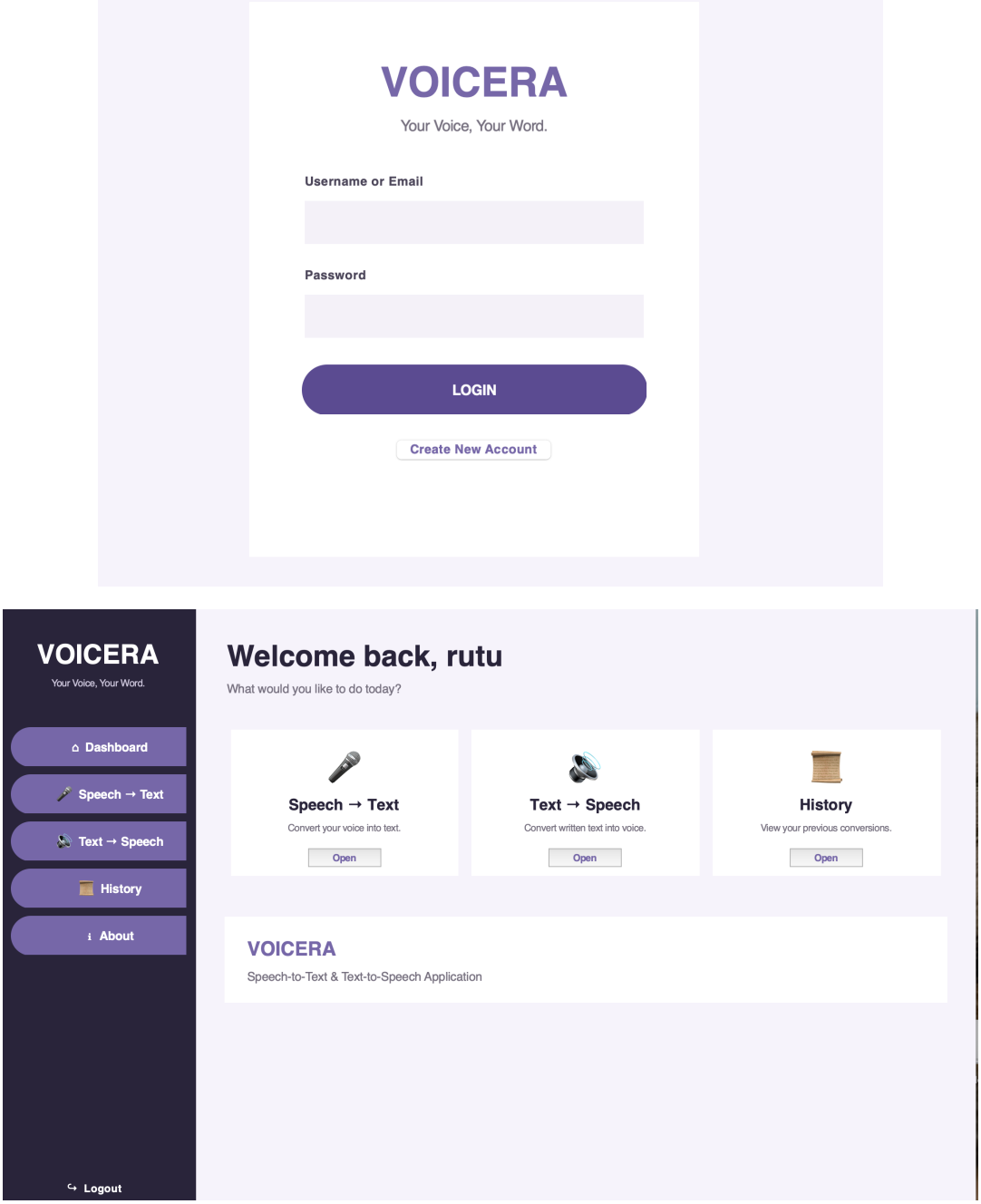


Figure 4.2: Login page and Dashboard

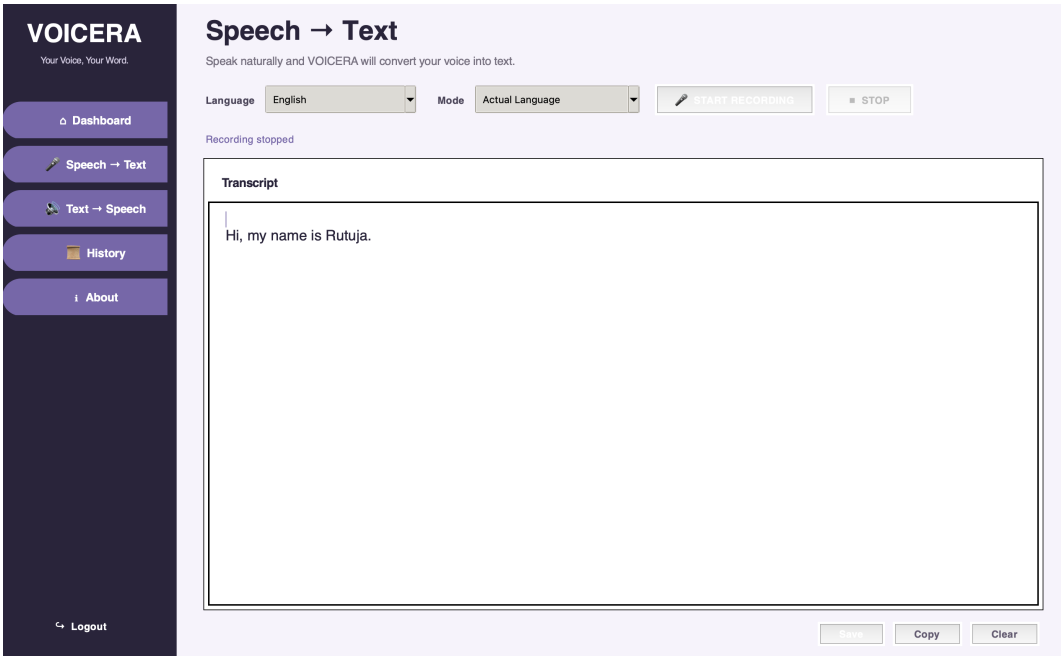
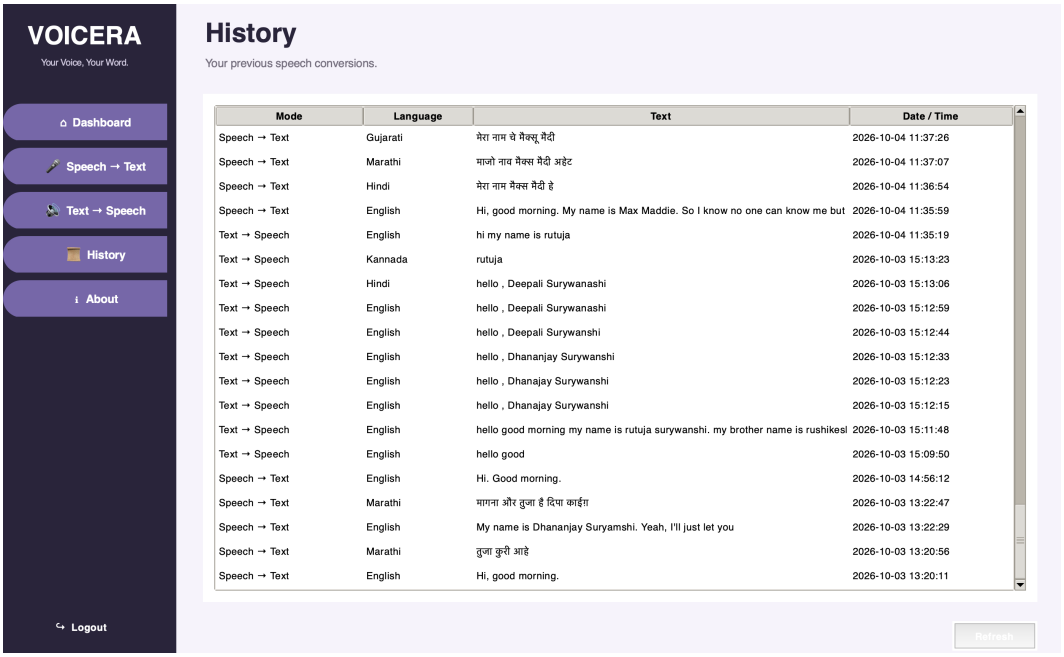


Figure 4.3: Output of the application



Mode	Language	Text	Date / Time
Speech → Text	Gujarati	मेरा नाम चे मैक्सू मैदी	2026-10-04 11:37:26
Speech → Text	Marathi	माजो नाव मैक्स मैदी अहेट	2026-10-04 11:37:07
Speech → Text	Hindi	मेरा नाम मैक्स मैदी है	2026-10-04 11:36:54
Speech → Text	English	Hi, good morning. My name is Max Maddie. So I know no one can know me but	2026-10-04 11:35:59
Text → Speech	English	hi my name is rutuja	2026-10-04 11:35:19
Text → Speech	Kannada	rutuja	2026-10-03 15:13:23
Text → Speech	Hindi	hello , Deepali Surywanashi	2026-10-03 15:13:06
Text → Speech	English	hello , Deepali Surywanashi	2026-10-03 15:12:59
Text → Speech	English	hello , Deepali Surywanshi	2026-10-03 15:12:44
Text → Speech	English	hello , Dhananjay Surywanshi	2026-10-03 15:12:33
Text → Speech	English	hello , Dhanajay Surywanshi	2026-10-03 15:12:23
Text → Speech	English	hello , Dhanajay Surywanshi	2026-10-03 15:12:15
Text → Speech	English	hello good morning my name is rutuja surywanshi. my brother name is rushikesl	2026-10-03 15:11:48
Text → Speech	English	hello good	2026-10-03 15:09:50
Speech → Text	English	Hi. Good morning.	2026-10-03 14:56:12
Speech → Text	Marathi	मागना और तुजा है दिपा काईग	2026-10-03 13:22:47
Speech → Text	English	My name is Dhananjay Suryamshi. Yeah, I'll just let you	2026-10-03 13:22:29
Speech → Text	Marathi	तुजा कुरी आहे	2026-10-03 13:20:56
Speech → Text	English	Hi, good morning.	2026-10-03 13:20:11

Figure 4.4: Listing tasks and marking task History

5. CONCLUSION

The VOICERA project shows how Python can be used to build a complete, working real-time multilingual speech-to-text and audio transcription application. Flask handles the web interface and the REST API, the intent engine understands commands using regular expressions, SQLite keeps notes, tasks and history, and the browser provides speech input and output. The application ran correctly for all tested commands, and all 9 unit tests passed. Arithmetic is evaluated safely with the ast module rather than eval().

Limitations: The intent engine is rule-based, so it understands only the supported command patterns, and voice input needs a browser that supports the Web Speech API (such as Chrome or Edge).

Future scope: The project can be extended with machine-learning based intent recognition, user login, weather and web search features, reminders with notifications, and support for more languages.